

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

eMethods.**Eligibility Criteria**

Inclusion criteria (patients meeting ALL the following criteria were enrolled):

- Age equal to or above 18 years.
- Both genders.
- In case of women, unwillingness to become pregnant during the study period.
- Written informed consent provided by the patient or by one first-degree relative/spouse in case of patients unable to consent.
- Community-acquired pneumonia (CAP) or hospital-acquired pneumonia (HAP) or ventilator-associated pneumonia (VAP) or primary bacteremia (BSI). These are defined in eTable 3 using three sets of criteria per infection. Enrolled patients should meet criteria as defined in eTable 3.
- Sepsis defined by the Sepsis-3 definitions. More precisely, sepsis is defined as the presence of total SOFA (sequential organ failure assessment) score equal to 2 or more for patients who are admitted with infection at the emergency department OR as any increase of admission SOFA by 2 or more points for patients already hospitalized.¹
- Patients with signs of fulminant hyper-inflammation (macrophage activation-like syndrome, MALS) or sepsis-induced immunoparalysis as defined in eTable 4. Since the state of MALS is considered more life-threatening than the state of immunoparalysis, patients with lab findings of both immune states are allocated to treatment targeting MALS. Patients with active COVID-19 infection according to the attending physicians may participate only in the MALS arm. Patients developing secondary infection in the field of COVID-19 according to the judgment of the attending physicians were allowed to be enrolled only if they had laboratory finding of sepsis-induced immunoparalysis (eTable 4)
- Time from classification into sepsis by the Sepsis-3 definitions and start of blind intervention less than 72 hours.

Exclusion criteria (patients meeting ANY of the following criteria COULD NOT be enrolled in the study):

- Age below 18 years.
- Denial for written informed consent.
- Acute pyelonephritis or intraabdominal infection, meningitis or skin infection.
- Any stage IV malignancy.
- Neutropenia defined as an absolute neutrophil count lower than $1,500/\text{mm}^3$.
- Any “do not resuscitate” decision in the hospital.
- In the case of BSI, patients with blood cultures growing coagulase-negative staphylococci or skin commensals or catheter-related infections could not be enrolled.
- Active tuberculosis (TB) as defined by the co-administration of drugs for the treatment of TB.
- Infection by the human immunodeficiency virus (HIV).
- Any primary immunodeficiency.
- Oral or intravenous intake of corticosteroids at a daily dose equal or greater than 0.4 mg/kg prednisone for more than the last 15 days.
- Any anti-cytokine biological treatment the last one month.
- Medical history of systemic lupus erythematosus.
- Medical history of multiple sclerosis or any other demyelinating disorder.
- Pregnancy or lactation. Women of child-bearing potential were screened by a urine pregnancy test before inclusion in the study.

Lab measurements

Ferritin was measured in serum by an enzyme immunosorbent assay (ORGENTEC Diagnostika GmbH, Mainz, Germany; lower detection limit 75ng/ml). Using whole blood collected in EDTA-coated vials, white blood cells were incubated for 15 minutes in the dark with the following monoclonal antibodies: anti-CD14 FITC, anti-HLA-DR-PE, anti-CD45 PC5 (Immunotech, Marseille, France) and Quantibrite HLA-DR/anti-monocyte PerCP-Cy5 (Quantibrite, BD Biosciences, Franklin Lakes, NJ, USA, CE-marked in vitro diagnostic (IVD) test). After incubation, cells were analysed by flow cytometer FC500 versus cells stained with anti-CD45 PC5 and anti-idiotypic IgG1. The number of antibodies bound on the surface antigen HLA-DR on CD45/CD14-positive monocytes was counted by the Navios EX flow cytometer (Beckman Coulter, Brea, USA) and considered an expression of the absolute count of HLA-DR receptors. Measurements of ferritin and HLA-DR of Greek study sites were done in the study central lab (Research Laboratory of Immunology, 4th Department of Internal Medicine, ATTIKON University General Hospital, Greece). Measurements of ferritin and HLA-DR in the other study sites were done following study lab protocols shared by the study sponsor (Hellenic Institute for the Study of Sepsis) and by using exactly the same reagents.

eTable 1. Total Number of Patients Screened and Randomized from each Study Site

Study site	Total Number of Screened Patients	Total Number of Randomized Patients
• 4th Department of Internal Medicine ATTIKON University Hospital, National and Kapodistrian University of Athens, Greece	64	23
• 2nd Department of Propaedeutic Internal Medicine ATTIKON University Hospital, National and Kapodistrian University of Athens, Greece	4	2
• 2nd Critical Care Department ATTIKON University Hospital, National and Kapodistrian University of Athens, Greece	19	13
• 1st Department of Pulmonary Medicine and Intensive Care Unit Sotiria General Hospital, Athens, Greece	1	0
• Department of Internal Medicine Larissa University Hospital, Larissa, Greece	5	1
• 2nd Intensive Care Unit, KAT Attica General Hospital, Kifissia, Greece	20	11
• Department of Critical Care Medicine, University of Ioannina, Ioannina, Greece	4	1
• Department of Critical Care Medicine, Democritus University of Thrace, Alexandroupolis, Greece	15	6
• Intensive Care Unit, G. Gennimatas General Hospital, Thessaloniki, Greece	21	4
• Intensive Care Unit, Aghios Dimitrios General Hospital, Thessaloniki, Greece	22	6
• Department of Anesthesiology and Intensive Care, AHEPA Thessaloniki University General Hospital, Thessaloniki, Greece	93	28
• Intensive Care Unit, Ippokrateion General Hospital of Thessaloniki, Thessaloniki, Greece	157	79

• Intensive Care Unit, Theageneion Anticancer Hospital, Thessaloniki, Greece	24	5
• Intensive Care Unit, Tzaneio General Hospital, Piraeus, Greece	1	1
• Intensive Care Unit, Koutlibaneio & Triantafylleio General Hospital, Larissa, Greece	1	1
• Intensive Care Unit of Latseion Burn Center, Thriasio General Hospital, Eleusis, Greece	12	3
• Department of Anesthesiology and Intensive Care, Jena University Hospital, Jena, Germany	11	9
• Departments of Intensive Care Medicine and Infectious Diseases, Amsterdam Medical University Center, Amsterdam, the Netherlands	24	5
• Intensive Care Unit and Department of Internal Medicine and Infectious Diseases, Radboud University Medical Center, Nijmegen, the Netherlands	17	5
• Department of Infectious Diseases, Clinical Hospital for Infectious Diseases Cluj-Napoca, Romania	4	2
• Service of Immunology and Allergy, Infectious Diseases and Adult Intensive Care Unit, Lausanne University Hospital, University of Lausanne, Lausanne, Switzerland	8	4
• Polyvalent Intensive Care Unit, Sotiria Athens General Hospital of Chest Diseases, Athens, Greece	37	20
• Intensive Care Unit, 424 Military General Hospital, Thessaloniki, Greece	2	2
• Intensive Care Unit, Asklepion General Hospital, Voula, Greece	32	14
• Intensive Care Unit of Center for Respiratory Failure, Sotiria Chest Diseases Athens General Hospital, Greece	28	17
• Intensive Care Unit, Korgialeneio-Benakeio Athens General Hospital, Athens, Greece	21	5

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|---|----|---|
| • Department of Intensive Care Medicine, University Hospital of Heraklion, University of Crete, Heraklion, Greece | 16 | 7 |
| • Clinic of Intensive Care and Pulmonary Diseases, Faculty of Nursing, School of Health Sciences, National and Kapodistrian University of Athens, Athens, Greece | 7 | 5 |
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eTable 2. List of Approvals from the Ethics Committees, Institution Review Boards or/and Central Regulatory Bodies for each Participating Country

Participating Country	Ethics Committee, Institution Review Board or/and Central Regulatory Body	Date of Approval	Number of Approval
Greece	National Organization for Medicines of Greece	18/02/2021	IS008/21
	National Ethics Committee of Greece	22/01/2021	02/2021
Germany	German Federal Institute for Drugs and Medical Devices	25/05/2022	61-3910-4045295
	Ethics Committee of the Jena University Hospital	02/03/2022	2022-2540-AMG-ff
The Netherlands	Central Committee on Research Involving Human Subjects for the Netherlands	03/06/2021	NL76706.091.21
	Ethics Committee of Radboud universitair medisch centrum	21/10/2021	PW/METC ON 512
Switzerland	Commission cantonale d'éthique de la recherche sur l'être humain (CER-VD)	01/06/2022	2022-00606
	Swiss Agency for Therapeutic Products, Swissmedic	23/05/2022	
Romania	National Agency for Medicine and Medical Products of Romania	26/09/2022	129E
	Comisia Nationala de Bioetica a Medicamentului si a Dispozitivelor Medicale	04/07/2022	19S1-4

Italy	Italian Medicines Agency – AIFA	15/02/2023	0019561- 15/02/2023-AIFA- AIFA_USC-P
	Direzione Scientifica Fondazione Policlinico Gemelli	28/03/2023	0010056/23

eTable 3. Definitions of Eligible Infections

	Criterion 1	Criterion 2	Criterion 3	Resolution criteria (to be considered for the secondary study endpoint)
Infection	All the following	At least 2 of the following	At least 1 of the following	
Primary bacteremia ²	- At least one positive blood culture - Failure to identify a primary infection site despite thorough clinical and radiology investigation	None	None	Sterile blood cultures after seven days
Community-acquired pneumonia ³	New infiltrate on chest X-ray	- New onset or worsening of cough - Dyspnea - Auscultatory findings compatible with pulmonary consolidation	- Procalcitonin ≥ 0.25 ng/ml - Hypoxemia with partial oxygen pressure ≤ 60 mmHg or oxygen saturation $\leq 90\%$ in ambient air - Respiratory rate ≥ 20 breaths/min	- None of criterion 2 findings making the patient eligible for the study AND - None of criterion 3 findings making the patient

				eligible for the study
Hospital-acquired pneumonia ⁴	- Onset >48 hours from hospital admission - New or evolving infiltrate on chest X-ray	- New onset or worsening of cough or dyspnea - Purulent tracheobronchial secretions - Auscultatory findings compatible with pulmonary consolidation	- Procalcitonin ≥ 0.25 ng/ml - Hypoxemia with partial oxygen pressure ≤ 60 mmHg or oxygen saturation $\leq 90\%$ in ambient air - Respiratory rate ≥ 20 breaths/min	- None of criterion 2 findings making the patient eligible for the study AND - None of criterion 3 findings making the patient eligible for the study
Ventilator-associated pneumonia ^{4,5}	- Onset >48 hours from start of mechanical ventilation - New or evolving infiltrate on chest X-ray	- Purulent tracheobronchial secretions - Auscultatory findings compatible with pulmonary consolidation	- Procalcitonin ≥ 0.25 ng/ml - Clinical pulmonary infection score (CPIS) ≥ 6	- None of criterion 2 findings making the patient eligible for the study AND - None of criterion 3 findings making the patient eligible for the study

Footnote: Any patient can be classified as suffering from one of the following infections if he/she meets listed criteria 1 to 3.

eTable 4. Combinations of Immunotests Guiding Randomization

	Macrophage activation- like syndrome		Unclassified	Sepsis-induced immunoparalysis
Ferritin (ng/ml)	>4,420	>4,420	≤4,420	≤4,420
HLA-DR (antibodies per CD45/CD14-monocyte)	≥5,000	<5,000	≥5,000	<5,000

Abbreviations: HLA-DR, Histocompatibility Complex DR.

eTable 5. Baseline Microbiology of Study Participants

	Standard care + Precision Immunotherapy (n=131)	Standard care + Placebo (n=145)
Baseline microbiology: blood cultures, No. (%)		
Gram-negative bacteria	19 (14.5)	16 (11.0)
<i>Acinetobacter baumannii</i>	5 (3.8)	9 (6.2)
<i>Klebsiella pneumoniae</i>	6 (4.6)	5 (3.4)
Others	8 (6.1)	2 (1.4)
Gram-positive bacteria	12 (9.2)	11 (7.6)
<i>Enterococcus faecium</i>	2 (1.5)	2 (1.4)
<i>Streptococcus pneumoniae</i>	0 (0.0)	2 (1.4)
<i>Staphylococcus aureus</i>	3 (2.3)	0 (0.0)
<i>Coagulase-negative Staphylococcus spp</i>	5 (3.8)	4 (2.8)
Others	2 (1.5)	3 (2.1)
<i>Candida spp</i>	1 (0.8)	3 (2.1)
Baseline microbiology: respiratory cultures*, No. (%)		
Gram-negative bacteria	47 (35.9)	48 (33.1)
<i>Acinetobacter baumannii</i>	14 (10.7)	22 (15.2)
<i>Klebsiella pneumoniae</i>	7 (5.3)	7 (4.8)
<i>Pseudomonas aeruginosa</i>	15 (11.5)	12 (8.3)
Others	11 (8.4)	7 (4.8)
Gram-positive bacteria	3 (2.3)	4 (2.8)
<i>Candida spp</i>	4 (3.1)	8 (5.5)

*isolated at $>10^5$ cfu/ml of tracheobronchial secretions

Abbreviations: n, number of patients.

eTable 6. Co-Administered Medication and Interventions

	Standard care + Precision Immunotherapy (n=131)	Standard care + Placebo (n=145)
Antibiotics, No. (%)		
Ampicillin/sulbactam	16 (12.2)	27 (18.6)
Piperacillin/tazobactam	27 (20.6)	25 (17.2)
3rd generation cephalosporins	12 (9.2)	14 (9.7)
Ceftaroline	6 (4.6)	4 (2.8)
Ceftazidime/avibactam	31 (23.7)	34 (23.4)
Ceftolozane/tazobactam	0 (0.0)	4 (2.8)
Ertapenem	3 (2.3)	3 (2.1)
Meropenem	47 (35.9)	43 (29.7)
Meropenem/varobactam	2 (1.5)	3 (2.1)
Tigecycline	27 (20.6)	25 (17.2)
Levofloxacin/Moxifloxacin	11 (8.4)	16 (11.0)
Colistin	74 (56.5)	84 (57.9)
Gentamycin	10 (7.6)	7 (4.8)
Amikacin	9 (6.9)	12 (8.3)
Fosfomycin	3 (2.3)	9 (6.2)
Linezolid	39 (29.8)	42 (29.0)
Glycopeptide (Vancomycin or Teicoplanin)	32 (24.4)	29 (20.0)
Daptomycin	37 (28.2)	40 (27.6)
Echinocandins	18 (13.7)	26 (17.9)
Treatment with at least one active antibiotic against the isolated pathogen, n/N (%) ^a	34/41 (82.9)	33/38 (86.8)
Other drugs/interventions, No. (%)		
Norepinephrine	110 (84.0)	126 (86.9)
Vasopressin	31 (23.7)	34 (23.4)
Mechanical ventilation	116 (88.5)	129 (89.0)
CVVH	23 (17.6)	24 (16.6)
LMWH	102 (77.9)	108 (74.5)
Low-dose hydrocortisone ^b	56 (42.7)	72 (49.7)

Dexamethasone	7 (5.3)	9 (6.2)
Methylprednisolone	21 (16.0)	20 (13.8)
Fast-acting insulin	44 (33.6)	50 (34.5)
Esmolol	16 (12.2)	24 (16.6)
Landiolol	9 (6.9)	8 (5.5)
Furosemide	82 (62.6)	95 (65.5)
Ipratropium	27 (20.6)	29 (20.0)
Ipratropium/salbutamol	43 (32.8)	41 (28.3)
Salbutamol	5 (3.8)	10 (6.9)

Abbreviations: CVVH, continuous veno-venal hemofiltration; LMWH, low molecular weight heparin; Q, quartile; SD, standard deviation.

^an refers to the number of patients treated with at least one active antibiotic against the isolated pathogen and N refers to the number of patients for whom antimicrobial susceptibility test was available.

^bLow-dose hydrocortisone indicates hydrocortisone at a dose of 200mg per day according to Surviving Sepsis Campaign Guidelines.⁶

eTable 7. Baseline Characteristics of Study Participants by State of Sepsis-Induced Immune Dysregulation

	Macrophage activation-like syndrome		Sepsis-induced immunoparalysis	
	Standard care + Anakinra (n=25)	Standard care + Placebo (n=23)	Standard care + rhIFN γ (n=106)	Standard care + Placebo (n=122)
Male, No. (%)	17 (68.0)	13 (56.5)	72 (67.9)	81 (66.4)
Female, No. (%)	8 (32.0)	10 (43.5)	34 (32.1)	41(33.6)
Age, mean (SD), years	66 (13)	66 (14)	70 (13)	70 (13)
Race, No. (%)				
African	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)
Caucasian	25 (100.0)	23 (100.0)	104 (98.1)	122 (100.0)
Middle Eastern	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)
Underlying Infection, No. (%)				
Ventilator-associated pneumonia	9 (36.0)	4 (17.4)	39 (36.8)	50 (41.0)
Hospital-acquired pneumonia	7 (28.0)	5 (21.7)	28 (26.4)	27 (22.1)
Community-acquired pneumonia	5 (20.0)	8 (34.8)	26 (24.5)	36 (29.5)
Primary bacteremia	3 (12.0)	3 (13.0)	13 (12.3)	9 (7.4)
Active SARS-CoV-2 pneumonia	1 (4.0)	3 (13.0)	0 (0.0)	0 (0.0)
APACHE II, median (Q1-Q3)	21 (17-26)	20 (16-26)	19 (15-25)	19 (14-25)
CCI, median (Q1-Q3)	3 (2-5)	4 (2-6)	4 (3-6)	5 (3-6)
SOFA, median (Q1-Q3)	10 (8-14)	10 (6-13.5)	9 (7-11)	9 (6-10)

Baseline laboratory values

Ferritin, median (Q1-Q3), ng/ml	5438 (5000-7009)	6221 (5000-7500)	633 (353-1149)	703 (340-1488)
mHLA-DR, median (Q1-Q3), antibodies per CD-14 cell	3576 (2124-6205)	4990 (3068-7811)	3043 (2297-4135)	3078 (1762-4004)
Total white blood cell count, mean (SD), /mm ³	16,834.0 (6,775.9)	18,597.4 (11,271.1)	17,044.2 (9,568.1)	16,585.4 (9,248.2)
Hemoglobin, mean (SD), g/dl	9.4 (1.8)	10.1 (1.9)	10.6 (7.6)	9.8 (1.8)
Platelets, mean (SD), /mm ³	233,480 (226,605)	142,683 (114,729)	193,729 (121,231)	227,385 (135,123)
International normalized ratio, median (Q1-Q3)	1.4 (1.1-1.5)	1.3 (1.1-1.5)	1.2 (1.1-1.3)	1.1 (1.0-1.3)
Fibrinogen, mean (SD), mg/dl	500.9 (258.9)	738.5 (950.7)	604.5 (220.3)	604.8 (922.2)
D-dimers, median (Q1-Q3), mg/dl	3.1 (0.6-8.9)	1.1 (0.6-3.4)	1.1 (0.5-2.9)	2.1 (0.5-5.9)
Creatinine, median (Q1-Q3), mg/dl	1.3 (0.8-2.5)	1.6 (0.9-2.8)	1.3 (0.8-1.9)	0.9 (0.7-1.5)
Creatinine 1.2-1.9 mg/dl, No. (%)	5 (20.0)	5 (21.7)	31 (29.2)	25 (20.5)
Creatinine 2.0-3.4 mg/dl, No. (%)	8 (32.0)	9 (39.1)	16 (15.1)	16 (13.1)
Creatinine 3.5-4.9 mg/dl, No. (%)	0 (0.0)	0 (0.0)	3 (2.8)	5 (4.1)
Creatinine ≥ 5.0 mg/dl, No. (%)	1 (4.0)	1 (4.3)	6 (5.7)	0 (0.0)
Total Bilirubin, median (Q1-Q3), mg/dl	0.9 (0.6-1.5)	0.5 (0.3-0.9)	0.6 (0.4-1.2)	0.6 (0.4-1.1)
Total bilirubin 1.2-1.9 mg/dl, No. (%)	4 (16.0)	2 (8.7)	12 (11.3)	14 (11.5)
Total bilirubin 2.0-5.9 mg/dl, No. (%)	4 (16.0)	2 (8.7)	10 (9.4)	11 (9.0)
Total bilirubin 6.0-11.9 mg/dl, No. (%)	1 (4.0)	0 (0.0)	1 (0.9)	0 (0.0)
Total bilirubin ≥ 12.0 mg/dl, No. (%)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)

Aspartate aminotransferase, median (Q1-Q3), U/l	184 (36-553)	71 (52-266)	39 (24-64)	37 (20-86)
Alanine aminotransaminase, median (Q1-Q3), U/l	187 (36-468)	46 (33-284)	32 (17-55)	33 (17-70)
pO ₂ /FiO ₂ , median (Q1-Q3)	143.1 (105.3-296.6)	159.7 (113.8-254.3)	167.5 (133.2-239.6)	172.0 (128.0-231.3)
Lactate, median (Q1-Q3), mmol/l	2.1 (1.3-5.9)	1.8 (1.2-3.6)	1.6 (1.1-2.3)	1.7 (1.0-3.0)
C-reactive protein, median (Q1-Q3), mg/l	18.0 (10.7-147.0)	41 (9.3-117.0)	31.3 (14.3-190.0)	32.0 (14.8-146.0)
Procalcitonin, median (Q1-Q3), ng/ml	3.4 (1.3-17.5)	1.8 (0.5-53.6)	1.1 (0.4-4.8)	1.1 (0.4-10.2)
Time between sepsis onset and start of the study drug, median (Q1-Q3), hours	34 (24-48)	48 (24-53)	38 (24-48)	46.5 (24-48)

Abbreviations: APACHE, acute physiology and chronic health evaluation; CCI, Chalron's comorbidity index; mHLA-DR, monocyctic histocompatibility complex DR; rhIFN γ , recombinant human activated interferon-gamma; Q, quartile; SARS-CoV-2, severe acute respiratory syndrome coronavirus-2; SD, standard deviation; SOFA, sequential organ failure assessment.

eTable 8. Cumulative Distribution of SOFA scores between Days 2 and 9

Each cell indicates on the respective day how many SOFA scores were actually measured; how many SOFA scores were given as 24 because of patients dying; and how many SOFA scores were assumed as last observations carried forward for patients discharged earlier from the ICU

Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Standard care + Precision Immunotherapy							
Measured= 130	Measured= 125	Measured= 119	Measured= 115	Measured= 114	Measured= 107	Measured= 102	Measured= 100
Assumed= 0	Assumed= 1	Assumed= 4	Assumed= 5	Assumed= 5	Assumed= 7	Assumed= 9	Assumed= 10
Death= 1	Death= 5	Death= 8	Death= 11	Death= 12	Death= 17	Death= 20	Death= 21
Standard care + Placebo							
Measured= 144	Measured= 135	Measured= 133	Measured= 127	Measured= 119	Measured= 117	Measured= 111	Measured= 103
Assumed= 0	Assumed= 2	Assumed= 2	Assumed= 4	Assumed= 7	Assumed= 8	Assumed= 10	Assumed= 13
Death= 1	Death= 8	Death= 10	Death= 14	Death= 19	Death= 20	Death= 24	Death= 29

Abbreviations: ICU, intensive care unit; SOFA, sequential organ failure assessment

eTable 9. Logistic Regression of Variables Impacting on the Primary Endpoint

	Meeting the endpoint		Univariate analysis		Multivariate analysis	
	No	Yes	OR	P value	OR	P value
Precision immuno-therapy, n/N (%)	85/204 (41.7)	46/72 (63.9)	2.48 (1.42 to 4.31)	.001	2.49 (1.42 to 4.36)	.001
APACHE II, median (Q1-Q3)	19 (14-25)	20 (16-25)	1.01 (0.97 to 1.05)	.650	1.00 (0.96 to 1.05)	.826
CCI, median (Q1-Q3)	5 (3-6)	4 (3-6)	0.94 (0.84 to 1.06)	.338	0.95 (0.84 to 1.07)	.375
SOFA, median (Q1-Q3)	9 (7-11)	9.5 (7-12.8)	1.03 (0.96 to 1.10)	.433	1.02 (0.95 to 1.09)	.562
Sites enrolling more than 8%, No. (%)	100 (49.0)	32 (44.4)	0.83 (0.48 to 1.42)	.504	0.80 (0.46 to 1.41)	.453

Abbreviations CCI, Charlson's comorbidity index; CI, confidence interval; n, number of patients; N, number of patients studied; OR odds ratio; Q, quartile, SOFA, sequential organ failure assessment.

eTable 10. Change of the mean SOFA score by Days 9 and 15 by the Level of Sepsis-Induced Immune Dysregulation

Endpoints	Standard care + Precision Immunotherapy	Standard care + Placebo	Difference, % (95% CIs)	OR _{unadjusted} (95% CI)	P value
≥1.4-point decrease of mean SOFA of days 2 to 9 from baseline day 1 per state of immune activation, n/N (%)					
Macrophage activation-like syndrome ^a	12/25 (48.0)	4/23 (17.4)	30.6 (3.9 to 51.8)	4.38 (1.15 to 16.64)	.034
Sepsis-induced immunoparalysis ^b	34/106 (32.1)	22/122 (18.0)	14.0 (2.8 to 25.1)	2.15 (1.16 to 3.97)	.020
≥1.4-point decrease of mean SOFA of days 2 to 15 from baseline day 1, n/N (%)					
Macrophage activation-like syndrome ^a (%)	12/25 (48.0)	5/23 (21.7)	26.3 (0.0 to 48.4)	3.32 (0.94 to 11.76)	.075
Sepsis-induced immunoparalysis ^b (%)	40/106 (37.7)	29/122 (23.8)	13.9 (1.9 to 25.6)	1.194 (1.09 to 3.45)	.030

Abbreviations CI, Confidence Interval; OR, Odds Ratio; n, number of patients; N, number of studied patients; NA, not applicable; SOFA, Sequential Organ Failure Assessment.

^adefined as blood ferritin >4,420 ng/ml.

^bdefined as ferritin ≤ 4,420 ng/ml and HLA-DR < 5,000 molecules on CD45/CD14-monocytes.

eTable 11. Comparison of Baseline Characteristics Between Patients With and Without Serial Measurements for the Secondary Endpoint of SIDF Reversal

	Reversal of SIDF analysis (n=125)	No Reversal of SIDF analysis (n=151)
Male, No. (%)	84 (67.2)	99 (65.6)
Female, No. (%)	41 (32.8)	52 (34.4)
Age, mean (SD), years	72 (12)	68 (14)
Most common comorbidities, No. (%)		
Diabetes mellitus	47 (37.6)	46 (30.5)
COVID-19 ^a	41 (32.8)	28 (18.5)
Coronary heart disease	33 (26.4)	26 (17.2)
Atrial fibrillation	27 (21.6)	32 (21.2)
Chronic obstructive pulmonary disease	25 (20.0)	40 (26.5)
Heart failure	18 (14.4)	23 (15.2)
Cerebral stroke	18 (14.4)	18 (11.9)
Chronic renal disease	12 (9.6)	17 (11.3)
Solid tumor malignancy without metastasis	9 (7.2)	12 (7.9)
Cerebral hemorrhage	7 (5.6)	11 (7.3)
Underlying Infection, No. (%)		
Ventilator-associated pneumonia	47 (37.6)	54 (35.8)
Hospital-acquired pneumonia	19 (15.2)	47 (31.1)
Community-acquired pneumonia	34 (27.2)	41 (27.2)
Primary bacteremia	19 (15.2)	9 (6.0)
Pneumonia by SARS-CoV-2	6 (4.8)	0 (0.0)
APACHE II, median (Q1-Q3)	19 (14-26)	20 (16-25)
CCI, median (Q1-Q3)	4 (3-6)	4 (3-5)
SOFA score, median (Q1-Q3)	9 (6-11)	10 (7-12)
Baseline laboratory values		
Ferritin at screening, median (Q1-Q3), ng/ml		
MALS	6170 (5425-7500)	5000 (4812-7598)
Immunoparalysis	678 (362-1140)	671 (337-1566)

mHLA-DR at screening, median (Q1-Q3), antibodies per CD-14 cell		
MALS	4425 (2130-6205)	4990 (2900-7811)
Immunoparalysis	3056 (1989-3962)	3075 (1932-4227)
Total white blood cell count, mean (SD), /mm ³	17,551.1 (10,352.4)	16,459.1 (8,399.7)
Hemoglobin, mean (SD), g/dl	9.96 (1.93)	10.21 (6.41)
Platelets, mean (SD), /mm ³	212,877 (128,703)	199,775 (152,901)
International normalized ratio, median (Q1-Q3)	1.2 (1.1-1.3)	1.2 (1.1-1.4)
Fibrinogen, mean (SD), mg/dl	584.4 (242.4)	625.8 (900.4)
D-dimers, median (Q1-Q3), mg/dl	717.0 (0.8-3432.7)	3.1 (0.7-1095.2)
Creatinine, median (Q1-Q3), mg/dl	1.0 (0.7-1.8)	1.2 (0.7-2.0)
Aspartate aminotransferase, median (Q1-Q3), U/l	37 (23-63)	53 (25-116)
Alanine aminotransaminase, median (Q1-Q3), U/l	33 (17-65)	40 (19-89)
Total bilirubin, median (Q1-Q3), mg/dl	0.6 (0.4-1.1)	0.6 (0.4-1.2)
pO ₂ /FiO ₂ , median (Q1-Q3)	164.2 (128.8-246.0)	173.1 (124.5-235.9)
Lactate, median (Q1-Q3), mmol/l	1.7 (1.10-2.60)	1.7 (1.10-3.10)
C-reactive protein, median (Q1-Q3), mg/l	35.0 (14.3-201.1)	30.1 (14.3-116.5)
Procalcitonin, median (Q1-Q3), ng/ml	1.3 (0.5-3.4)	1.3 (0.5-13.8)
Time between sepsis onset and start of the study drug, median (Q1-Q3), hours	30.0 (24.0-48.0)	48.0 (24.0-50.0)

Abbreviations: APACHE, Acute Physiology and Chronic Health Evaluation; CCI, Charlson's comorbidity index; mHLA-DR, monocytic histocompatibility complex DR; MALS, macrophage activation-like syndrome; Q, quartile; SARS-CoV-2, severe acute respiratory syndrome coronavirus-2; SD, standard deviation; SIDF, sepsis-induced immune dysfunction; SOFA, sequential organ failure assessment.

^aRefers to history of resolved COVID-19 infection.

**eTable 12. Suspected Unexpected Serious Adverse Reactions (SUSARs) by System-Organ Class and Preferred Term
Classified by Expectancy (Safety Set)**

	Macrophage Activation-like Syndrome (N= 48)		Sepsis-Induced Immunoparalysis (N= 228)		
	Standard care + Anakinra (N=25)	Standard care + Placebo (N=23)	Standard care + rhIFNγ (N=106)	Standard care + Placebo (N=122)	Total (N=276)
Any SUSAR, n (%)^a	0 (0.0)	0 (0.0)	2 (1.9)	3 (2.5)	5 (1.8)
Unexpected	0 (0.0)	0 (0.0)	2 (1.9)	3 (2.5)	5 (1.8)
Cardiac disorders, n (%)^a	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Cardiac arrest, n (%)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Unexpected	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
General disorders and administration site conditions, n (%)^a	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Multiple organ dysfunction syndrome, n (%)^b	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Unexpected	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Hepatobiliary disorders, n (%)^a	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Liver injury, n (%)	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Unexpected	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Infections and infestations, n (%)^a	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Septic shock, n (%)^c	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)

Unexpected	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Respiratory, thoracic and mediastinal disorders, n (%)^a	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Mediastinal haemorrhage, n (%)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Unexpected	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Vascular disorders, n (%)^a	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Hypotension, n (%)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Unexpected	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)

Abbreviations: PT, preferred term; rhIFN γ , recombinant human activated interferon-gamma; SOC, system organ class; SUSAR, suspected unexpected serious adverse reaction.

Footnote: The SUSARs were classified by expectancy using Medra 27.0 for events captured between days 1 and 90 of follow-up for the 276 patients. It is clarified that the total of each SOC refers to the number of patients and not to the cumulative number of events.

^aThese data represent the number of patients, whereas all other data in the table correspond to the number of events.

^bRepresents the progression to Multiple organ dysfunction syndrome (MODS) of the initial sepsis or septic shock for which the patient was enrolled in the study.

^cRepresents a new episode of sepsis or septic shock after the resolution of the initial sepsis or septic shock for which the patient was enrolled in the study.

eTable 13. Serious TEAEs (>2% Incidence) by System-Organ Class and Preferred Term; Total Events Classified by Relationship to Study Drug (Safety set)

	Macrophage Activation-like Syndrome (N= 48)		Sepsis-Induced Immunoparalysis (N= 228)		
	Standard care + Anakinra (N=25)	Standard care + Placebo (N=23)	Standard care + rhIFNγ (N=106)	Standard care + Placebo (N=122)	Total (N=276)
Any SAE, n (%)^a	20 (80.0)	21 (91.3)	96 (90.6)	108 (88.5)	245 (88.8)
Probably related	1 (4.0)	0 (0.0)	1 (0.9)	1 (0.8)	3 (1.1)
Possibly related	1 (4.0)	0 (0.0)	5 (4.7)	2 (1.6)	8 (2.9)
Probably not related	4 (16.0)	9 (39.1)	28 (26.4)	26 (21.3)	67 (24.3)
Unrelated	14 (56.0)	12 (52.2)	62 (58.5)	79 (64.8)	167 (60.5)
Blood and lymphatic system disorders, n (%)^a	13 (52.0)	4 (17.4)	40 (37.7)	38 (31.1)	95 (34.4)
Anaemia, n (%)	12 (48.0)	4 (17.4)	35 (33.0)	34 (27.9)	85 (30.8)
Thrombocytopenia, n (%)	2 (8.0)	2 (8.7)	12 (11.3)	7 (5.7)	23 (8.3)
Cardiac disorders, n (%)^a	3 (12.0)	6 (26.1)	31 (29.2)	29 (23.8)	69 (25.0)
Atrial fibrillation, n (%)	2 (8.0)	4 (17.4)	17 (16.0)	10 (8.2)	33 (12.0)
Bradycardia, n (%)	1 (4.0)	2 (8.7)	5 (4.7)	4 (3.3)	12 (4.3)
Cardiac arrest, n (%)	1 (4.0)	3 (13.0)	8 (7.5)	13 (10.7)	25 (9.1)
General disorders and administration site conditions, n (%)^a	10 (40.0)	11 (47.8)	46 (43.4)	51 (41.8)	118 (42.8)

Multiple organ dysfunction syndrome, n (%)^b	9 (36.0)	10 (43.5)	44 (41.5)	48 (39.3)	111 (40.2)
Infections and infestations, n (%)^a	11 (44.0)	9 (39.1)	59 (55.7)	56 (45.9)	135 (48.9)
Bacteraemia, n (%)	6 (24.0)	6 (26.1)	30 (28.3)	21 (17.2)	63 (22.8)
Device related infection, n (%)	0 (0.0)	2 (8.7)	2 (1.9)	2 (1.6)	6 (2.2)
Fungaemia, n (%)	2 (8.0)	2 (8.7)	15 (14.2)	10 (8.2)	29 (10.5)
Fungal infection, n (%)	1 (4.0)	0 (0.0)	6 (5.7)	5 (4.1)	12 (4.3)
Pneumonia, n (%)	5 (20.0)	6 (26.1)	30 (28.3)	26 (21.3)	67 (24.3)
Sepsis, n (%)^c	3 (12.0)	2 (8.7)	6 (5.7)	10 (8.2)	21 (7.6)
Septic shock, n (%)^c	7 (28.0)	3 (13.0)	22 (20.8)	14 (11.5)	46 (16.7)
Urinary tract infection, n (%)	1 (4.0)	0 (0.0)	3 (2.8)	7 (5.7)	11 (4.0)
Renal and urinary disorders, n (%)^a	4 (16.0)	4 (17.4)	19 (17.9)	16 (13.1)	43 (15.6)
Acute kidney injury, n (%)	3 (12.0)	3 (13.0)	18 (17.0)	16 (13.1)	40 (14.5)
Respiratory, thoracic and mediastinal disorders, n (%)^a	4 (16.0)	4 (17.4)	27 (25.5)	24 (19.7)	59 (21.4)
Bronchial obstruction, n (%)	1 (4.0)	1 (4.3)	2 (1.9)	2 (1.6)	6 (2.2)
Hypoxia, n (%)	1 (4.0)	1 (4.3)	3 (2.8)	6 (4.9)	11 (4.0)
Pleural effusion, n (%)	0 (0.0)	2 (8.7)	5 (4.7)	1 (0.8)	8 (2.9)
Pneumothorax, n (%)	3 (12.0)	0 (0.0)	7 (6.6)	7 (5.7)	17 (6.2)
Respiratory failure, n (%)	0 (0.0)	0 (0.0)	2 (1.9)	7 (5.7)	9 (3.3)
Tracheal stenosis, n (%)	1 (4.0)	1 (4.3)	6 (5.7)	7 (5.7)	15 (5.4)

Vascular disorders, n (%)^a	2 (8.0)	3 (13.0)	8 (7.5)	7 (5.7)	20 (7.2)
Haemorrhage, n (%)	1 (4.0)	0 (0.0)	5 (4.7)	0 (0.0)	6 (2.2)

Abbreviations: MALS, macrophage activation-like syndrome; PT, preferred term; rhIFN γ , recombinant human activated interferon-gamma; SOC, system organ class; TEAE, treatment-emergent adverse event.

Footnote: The serious TEAEs (sTEAEs) were classified by relationship to the study drug using Medra 27.0 for events captured during the 90 days of follow-up for the enrolled 276 patients. In the analysis, sTEAEs are graded as probably related, possibly related, probably not related, and unrelated. It is clarified that the total of each SOC refers to the number of patients and not to the cumulative number of events.

^aThese data represent the number of patients, whereas all other data in the table correspond to the number of events.

^bRepresents the progression to Multiple organ dysfunction syndrome (MODS) of the initial sepsis or septic shock for which the patient was enrolled in the study.

^cRepresents a new episode of sepsis or septic shock after the resolution of the initial sepsis or septic shock for which the patient was enrolled in the study.

eTable 14. Non-Serious TEAEs (>2% Incidence) by System-Organ Class and Preferred Term Classified by Maximum Severity; Total Events Classified by Relationship to Study Drug and by Maximum Severity (Safety Set)

	Macrophage Activation-like Syndrome (N= 48)		Sepsis-Induced Immunoparalysis (N= 228)		
	Standard care + Anakinra (N=25)	Standard care + Placebo (N=23)	Standard care + rhIFNγ (N=106)	Standard care + Placebo (N=122)	Total (N=276)
Any AE; Classification by Relationship to Study Drug, n (%)^a	20 (80.0)	20 (87.0)	104 (98.1)	112 (91.8)	256 (92.8)
Probably related	0 0 (0.0)	1 (4.3)	3 (2.8)	1 (0.8)	5 (1.8)
Possibly related	2 (8.0)	1 (4.3)	8 (7.5)	5 (4.1)	16 (5.8)
Probably not related	13 (52.0)	11 (47.8)	70 (66.0)	77 (63.1)	171 (62.0)
Unrelated	5 (20.0)	7 (30.4)	23 (21.7)	29 (23.8)	64 (23.2)
Any AE; Classification by Maximum Severity, n (%)^a	20 (80.0)	20 (87.0)	104 (98.1)	112 (91.8)	256 (92.8)
Mild	2 (8.0)	1 (4.3)	19 (17.9)	25 (20.5)	47 (17.0)
Moderate	6 (24.0)	5 (21.7)	40 (37.7)	42 (34.4)	93 (33.7)
Severe	12 (48.0)	14 (60.9)	45 (42.5)	45 (36.9)	116 (42.0)
Blood and lymphatic system disorders, n (%)^a	15 (60.0)	14 (60.9)	61 (57.5)	65 (53.3)	155 (56.2)
Anaemia	9 (36.0)	9 (39.1)	44 (41.5)	47 (38.5)	109 (39.5)
Mild	0 (0.0)	2 (8.7)	11 (10.4)	11 (9.0)	24 (8.7)

Moderate	2 (8.0)	2 (8.7)	13 (12.3)	14 (11.5)	31 (11.2)
Severe	7 (28.0)	5 (21.7)	20 (18.9)	22 (18.0)	54 (19.6)
Leukocytosis	5 (20.0)	4 (17.4)	19 (17.9)	19 (15.6)	47 (17.0)
Mild	1 (4.0)	2 (8.7)	7 (6.6)	9 (7.4)	19 (6.9)
Moderate	3 (12.0)	2 (8.7)	9 (8.5)	10 (8.2)	24 (8.7)
Severe	1 (4.0)	0 (0.0)	3 (2.8)	0 (0.0)	4 (1.4)
Cardiac disorders^a	3 (12.0)	5 (21.7)	19 (17.9)	27 (22.1)	54 (19.6)
Atrial fibrillation	1 (4.0)	2 (8.7)	7 (6.6)	8 (6.6)	18 (6.5)
Mild	0 (0.0)	0 (0.0)	0 (0.0)	3 (2.5)	3 (1.1)
Moderate	1 (4.0)	1 (4.3)	2 (1.9)	4 (3.3)	8 (2.9)
Severe	0 (0.0)	1 (4.3)	5 (4.7)	1 (0.8)	7 (2.5)
Bradycardia	0 (0.0)	2 (8.7)	1 (0.9)	9 (7.4)	12 (4.3)
Mild	0 (0.0)	1 (4.3)	0 (0.0)	7 (5.7)	8 (2.9)
Moderate	0 (0.0)	1 (4.3)	1 (0.9)	1 (0.8)	3 (1.1)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Tachycardia	1 (4.0)	0 (0.0)	10 (9.4)	10 (8.2)	21 (7.6)
Mild	0 (0.0)	0 (0.0)	6 (5.7)	9 (7.4)	15 (5.4)
Moderate	1 (4.0)	0 (0.0)	4 (3.8)	1 (0.8)	6 (2.2)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Gastrointestinal disorders^a	3 (12.0)	5 (21.7)	30 (28.3)	30 (24.6)	68 (24.6)
Constipation	0 (0.0)	1 (4.3)	7 (6.6)	10 (8.2)	18 (6.5)

Mild	0 (0.0)	1 (4.3)	6 (5.7)	9 (7.4)	16 (5.8)
Moderate	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Diarrhoea	3 (12.0)	2 (8.7)	20 (18.9)	19 (15.6)	44 (15.9)
Mild	2 (8.0)	2 (8.7)	18 (17.0)	17 (13.9)	39 (14.1)
Moderate	1 (4.0)	0 (0.0)	2 (1.9)	1 (0.8)	4 (1.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Ileus	0 (0.0)	4 (17.4)	2 (1.9)	3 (2.5)	9 (3.3)
Mild	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Moderate	0 (0.0)	4 (17.4)	2 (1.9)	1 (0.8)	7 (2.5)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Vomiting	0 (0.0)	0 (0.0)	5 (4.7)	5 (4.1)	10 (3.6)
Mild	0 (0.0)	0 (0.0)	5 (4.7)	4 (3.3)	9 (3.3)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
General disorders and administration site conditions, n (%)^a	3 (12.0)	2 (8.7)	25 (23.6)	29 (23.8)	59 (21.4)
Hypothermia	1 (4.0)	1 (4.3)	5 (4.7)	6 (4.9)	13 (4.7)
Mild	1 (4.0)	1 (4.3)	4 (3.8)	5 (4.1)	11 (4.0)
Moderate	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)

Pyrexia	2 (8.0)	1 (4.3)	20 (18.9)	23 (18.9)	46 (16.7)
Mild	2 (8.0)	1 (4.3)	16 (15.1)	18 (14.8)	37 (13.4)
Moderate	0 (0.0)	0 (0.0)	4 (3.8)	5 (4.1)	9 (3.3)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Infections and infestations, n (%)^a	3 (12.0)	3 (13.0)	18 (17.0)	21 (17.2)	45 (16.3)
Fungal infection	1 (4.0)	0 (0.0)	7 (6.6)	9 (7.4)	17 (6.2)
Mild	1 (4.0)	0 (0.0)	7 (6.6)	9 (7.4)	17 (6.2)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Urinary tract infection	1 (4.0)	0 (0.0)	4 (3.8)	4 (3.3)	9 (3.3)
Mild	0 (0.0)	0 (0.0)	2 (1.9)	3 (2.5)	5 (1.8)
Moderate	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)
Severe	1 (4.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.4)
Injury, poisoning and procedural complications, n (%)^a	1 (4.0)	0 (0.0)	2 (1.9)	6 (4.9)	9 (3.3)
Vascular access complication	1 (4.0)	0 (0.0)	2 (1.9)	5 (4.1)	8 (2.9)
Mild	1 (4.0)	0 (0.0)	1 (0.9)	3 (2.5)	5 (1.8)
Moderate	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Investigations, n (%)^a	18 (72.0)	17 (73.9)	86 (81.1)	92 (75.4)	213 (77.2)

Activated partial thromboplastin time prolonged	0 (0.0)	2 (8.7)	3 (2.8)	6 (4.9)	11 (4.0)
Mild	0 (0.0)	0 (0.0)	0 (0.0)	3 (2.5)	3 (1.1)
Moderate	0 (0.0)	1 (4.3)	2 (1.9)	2 (1.6)	5 (1.8)
Severe	0 (0.0)	1 (4.3)	1 (0.9)	1 (0.8)	3 (1.1)
Alanine aminotransferase increased	7 (28.0)	9 (39.1)	42 (39.6)	36 (29.5)	94 (34.1)
Mild	3 (12.0)	3 (13.0)	27 (25.5)	23 (18.9)	56 (20.3)
Moderate	3 (12.0)	3 (13.0)	8 (7.5)	9 (7.4)	23 (8.3)
Severe	1 (4.0)	3 (13.0)	7 (6.6)	4 (3.3)	15 (5.4)
Amylase increased	4 (16.0)	7 (30.4)	22 (20.8)	22 (18.0)	55 (19.9)
Mild	2 (8.0)	4 (17.4)	16 (15.1)	16 (13.1)	38 (13.8)
Moderate	2 (8.0)	3 (13.0)	5 (4.7)	3 (2.5)	13 (4.7)
Severe	0 (0.0)	0 (0.0)	1 (0.9)	3 (2.5)	4 (1.4)
Aspartate aminotransferase increased	9 (36.0)	10 (43.5)	54 (50.9)	47 (38.5)	120 (43.5)
Mild	3 (12.0)	5 (21.7)	30 (28.3)	30 (24.6)	68 (24.6)
Moderate	5 (20.0)	3 (13.0)	15 (14.2)	13 (10.7)	36 (13.0)
Severe	1 (4.0)	2 (8.7)	9 (8.5)	4 (3.3)	16 (5.8)
Blood alkaline phosphatase increased	8 (32.0)	7 (30.4)	28 (26.4)	42 (34.4)	85 (30.8)
Mild	5 (20.0)	5 (21.7)	22 (20.8)	29 (23.8)	61 (22.1)
Moderate	2 (8.0)	0 (0.0)	4 (3.8)	6 (4.9)	12 (4.3)
Severe	1 (4.0)	2 (8.7)	2 (1.9)	7 (5.7)	12 (4.3)

Blood bilirubin increased	4 (16.0)	7 (30.4)	22 (20.8)	23 (18.9)	56 (20.3)
Mild	3 (12.0)	3 (13.0)	9 (8.5)	12 (9.8)	27 (9.8)
Moderate	0 (0.0)	3 (13.0)	9 (8.5)	5 (4.1)	17 (6.2)
Severe	1 (4.0)	1 (4.3)	4 (3.8)	6 (4.9)	12 (4.3)
Blood creatine phosphokinase increased	1 (4.0)	3 (13.0)	14 (13.2)	13 (10.7)	31 (11.2)
Mild	1 (4.0)	3 (13.0)	11 (10.4)	12 (9.8)	27 (9.8)
Moderate	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)
Severe	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Blood creatinine increased	8 (32.0)	8 (34.8)	32 (30.2)	22 (18.0)	70 (25.4)
Mild	3 (12.0)	5 (21.7)	13 (12.3)	12 (9.8)	33 (12.0)
Moderate	4 (16.0)	1 (4.3)	17 (16.0)	9 (7.4)	31 (11.2)
Severe	1 (4.0)	2 (8.7)	2 (1.9)	1 (0.8)	6 (2.2)
Blood fibrinogen decreased	0 (0.0)	0 (0.0)	7 (6.6)	1 (0.8)	8 (2.9)
Mild	0 (0.0)	0 (0.0)	4 (3.8)	0 (0.0)	4 (1.4)
Moderate	0 (0.0)	0 (0.0)	3 (2.8)	0 (0.0)	3 (1.1)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Blood lactate dehydrogenase increased	1 (4.0)	3 (13.0)	13 (12.3)	13 (10.7)	30 (10.9)
Mild	1 (4.0)	2 (8.7)	10 (9.4)	12 (9.8)	25 (9.1)
Moderate	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)
Severe	0 (0.0)	1 (4.3)	1 (0.9)	0 (0.0)	2 (0.7)
Blood urea increased	0 (0.0)	0 (0.0)	6 (5.7)	6 (4.9)	12 (4.3)

Mild	0 (0.0)	0 (0.0)	5 (4.7)	5 (4.1)	10 (3.6)
Moderate	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Cortisol decreased	1 (4.0)	1 (4.3)	2 (1.9)	4 (3.3)	8 (2.9)
Mild	0 (0.0)	1 (4.3)	2 (1.9)	3 (2.5)	6 (2.2)
Moderate	1 (4.0)	0 (0.0)	0 (0.0)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Creatinine renal clearance decreased	0 (0.0)	1 (4.3)	3 (2.8)	2 (1.6)	6 (2.2)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Moderate	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Severe	0 (0.0)	1 (4.3)	1 (0.9)	1 (0.8)	3 (1.1)
Gamma-glutamyltransferase increased	9 (36.0)	10 (43.5)	39 (36.8)	29 (23.8)	87 (31.5)
Mild	5 (20.0)	4 (17.4)	24 (22.6)	15 (12.3)	48 (17.4)
Moderate	1 (4.0)	6 (26.1)	13 (12.3)	9 (7.4)	29 (10.5)
Severe	3 (12.0)	0 (0.0)	2 (1.9)	5 (4.1)	10 (3.6)
Hepatic enzyme increased	1 (4.0)	0 (0.0)	7 (6.6)	6 (4.9)	14 (5.1)
Mild	1 (4.0)	0 (0.0)	3 (2.8)	3 (2.5)	7 (2.5)
Moderate	0 (0.0)	0 (0.0)	3 (2.8)	2 (1.6)	5 (1.8)
Severe	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
International normalised ratio increased	1 (4.0)	5 (21.7)	19 (17.9)	21 (17.2)	46 (16.7)
Mild	0 (0.0)	4 (17.4)	12 (11.3)	13 (10.7)	29 (10.5)

Moderate	0 (0.0)	1 (4.3)	7 (6.6)	6 (4.9)	14 (5.1)
Severe	1 (4.0)	0 (0.0)	0 (0.0)	2 (1.6)	3 (1.1)
Lymphocyte count decreased	1 (4.0)	1 (4.3)	4 (3.8)	3 (2.5)	9 (3.3)
Mild	1 (4.0)	1 (4.3)	2 (1.9)	1 (0.8)	5 (1.8)
Moderate	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Platelet count decreased	9 (36.0)	13 (56.5)	42 (39.6)	49 (40.2)	113 (40.9)
Mild	3 (12.0)	4 (17.4)	16 (15.1)	30 (24.6)	53 (19.2)
Moderate	2 (8.0)	4 (17.4)	13 (12.3)	9 (7.4)	28 (10.1)
Severe	4 (16.0)	5 (21.7)	13 (12.3)	10 (8.2)	32 (11.6)
Platelet count increased	0 (0.0)	1 (4.3)	4 (3.8)	4 (3.3)	9 (3.3)
Mild	0 (0.0)	1 (4.3)	4 (3.8)	3 (2.5)	8 (2.9)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Troponin increased	2 (8.0)	1 (4.3)	5 (4.7)	6 (4.9)	14 (5.1)
Mild	1 (4.0)	0 (0.0)	1 (0.9)	5 (4.1)	7 (2.5)
Moderate	0 (0.0)	1 (4.3)	1 (0.9)	0 (0.0)	2 (0.7)
Severe	1 (4.0)	0 (0.0)	3 (2.8)	1 (0.8)	5 (1.8)
Urine output decreased	0 (0.0)	1 (4.3)	2 (1.9)	5 (4.1)	8 (2.9)
Mild	0 (0.0)	0 (0.0)	2 (1.9)	3 (2.5)	5 (1.8)
Moderate	0 (0.0)	1 (4.3)	0 (0.0)	0 (0.0)	1 (0.4)

Severe	0 (0.0)	0 (0.0)	0 (0.0)	2 (1.6)	2 (0.7)
White blood cell count decreased	3 (12.0)	0 (0.0)	12 (11.3)	9 (7.4)	24 (8.7)
Mild	1 (4.0)	0 (0.0)	9 (8.5)	9 (7.4)	19 (6.9)
Moderate	2 (8.0)	0 (0.0)	3 (2.8)	0 (0.0)	5 (1.8)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Metabolism and nutrition disorders, n (%)^a	12 (48.0)	15 (65.2)	64 (60.4)	65 (53.3)	156 (56.5)
Acidosis	0 (0.0)	1 (4.3)	3 (2.8)	6 (4.9)	10 (3.6)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	1 (0.8)	2 (0.7)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	0 (0.0)	1 (4.3)	2 (1.9)	4 (3.3)	7 (2.5)
Alkalosis	0 (0.0)	2 (8.7)	4 (3.8)	5 (4.1)	11 (4.0)
Mild	0 (0.0)	0 (0.0)	0 (0.0)	2 (1.6)	2 (0.7)
Moderate	0 (0.0)	1 (4.3)	0 (0.0)	0 (0.0)	1 (0.4)
Severe	0 (0.0)	1 (4.3)	4 (3.8)	3 (2.5)	8 (2.9)
Hyperglycaemia	3 (12.0)	3 (13.0)	11 (10.4)	19 (15.6)	36 (13.0)
Mild	2 (8.0)	3 (13.0)	8 (7.5)	15 (12.3)	28 (10.1)
Moderate	1 (4.0)	0 (0.0)	0 (0.0)	4 (3.3)	5 (1.8)
Severe	0 (0.0)	0 (0.0)	3 (2.8)	0 (0.0)	3 (1.1)
Hyperkalaemia	4 (16.0)	4 (17.4)	17 (16.0)	16 (13.1)	41 (14.9)
Mild	2 (8.0)	1 (4.3)	8 (7.5)	13 (10.7)	24 (8.7)
Moderate	2 (8.0)	1 (4.3)	5 (4.7)	3 (2.5)	11 (4.0)

Severe	0 (0.0)	2 (8.7)	4 (3.8)	0 (0.0)	6 (2.2)
Hypermagnesaemia	1 (4.0)	4 (17.4)	9 (8.5)	5 (4.1)	19 (6.9)
Mild	0 (0.0)	3 (13.0)	5 (4.7)	5 (4.1)	13 (4.7)
Moderate	1 (4.0)	0 (0.0)	1 (0.9)	0 (0.0)	2 (0.7)
Severe	0 (0.0)	1 (4.3)	3 (2.8)	0 (0.0)	4 (1.4)
Hypernatraemia	7 (28.0)	10 (43.5)	37 (34.9)	36 (29.5)	90 (32.6)
Mild	4 (16.0)	5 (21.7)	23 (21.7)	24 (19.7)	56 (20.3)
Moderate	0 (0.0)	4 (17.4)	10 (9.4)	10 (8.2)	24 (8.7)
Severe	3 (12.0)	1 (4.3)	4 (3.8)	2 (1.6)	10 (3.6)
Hyperphosphataemia	1 (4.0)	4 (17.4)	6 (5.7)	6 (4.9)	17 (6.2)
Mild	1 (4.0)	3 (13.0)	6 (5.7)	6 (4.9)	16 (5.8)
Moderate	0 (0.0)	1 (4.3)	0 (0.0)	0 (0.0)	1 (0.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Hypertriglyceridaemia	2 (8.0)	0 (0.0)	2 (1.9)	4 (3.3)	8 (2.9)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	3 (2.5)	4 (1.4)
Moderate	1 (4.0)	0 (0.0)	1 (0.9)	1 (0.8)	3 (1.1)
Severe	1 (4.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.4)
Hypoalbuminaemia	3 (12.0)	4 (17.4)	15 (14.2)	14 (11.5)	36 (13.0)
Mild	0 (0.0)	0 (0.0)	3 (2.8)	4 (3.3)	7 (2.5)
Moderate	3 (12.0)	2 (8.7)	9 (8.5)	8 (6.6)	22 (8.0)
Severe	0 (0.0)	2 (8.7)	3 (2.8)	2 (1.6)	7 (2.5)

Hypocalcaemia	2 (8.0)	2 (8.7)	15 (14.2)	8 (6.6)	27 (9.8)
Mild	1 (4.0)	0 (0.0)	6 (5.7)	6 (4.9)	13 (4.7)
Moderate	0 (0.0)	1 (4.3)	6 (5.7)	1 (0.8)	8 (2.9)
Severe	1 (4.0)	1 (4.3)	3 (2.8)	1 (0.8)	6 (2.2)
Hypoglycaemia	4 (16.0)	4 (17.4)	15 (14.2)	17 (13.9)	40 (14.5)
Mild	2 (8.0)	3 (13.0)	9 (8.5)	14 (11.5)	28 (10.1)
Moderate	2 (8.0)	1 (4.3)	6 (5.7)	3 (2.5)	12 (4.3)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Hypokalaemia	2 (8.0)	3 (13.0)	23 (21.7)	24 (19.7)	52 (18.8)
Mild	2 (8.0)	1 (4.3)	12 (11.3)	21 (17.2)	36 (13.0)
Moderate	0 (0.0)	1 (4.3)	7 (6.6)	1 (0.8)	9 (3.3)
Severe	0 (0.0)	1 (4.3)	4 (3.8)	2 (1.6)	7 (2.5)
Hypomagnesaemia	1 (4.0)	3 (13.0)	8 (7.5)	8 (6.6)	20 (7.2)
Mild	1 (4.0)	3 (13.0)	8 (7.5)	7 (5.7)	19 (6.9)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Hyponatraemia	2 (8.0)	3 (13.0)	17 (16.0)	11 (9.0)	33 (12.0)
Mild	2 (8.0)	3 (13.0)	17 (16.0)	10 (8.2)	32 (11.6)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Hypophosphataemia	1 (4.0)	4 (17.4)	11 (10.4)	12 (9.8)	28 (10.1)

Mild	0 (0.0)	2 (8.7)	8 (7.5)	9 (7.4)	19 (6.9)
Moderate	1 (4.0)	2 (8.7)	3 (2.8)	3 (2.5)	9 (3.3)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Metabolic acidosis	1 (4.0)	1 (4.3)	2 (1.9)	2 (1.6)	6 (2.2)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	1 (0.4)
Moderate	0 (0.0)	1 (4.3)	0 (0.0)	1 (0.8)	2 (0.7)
Severe	1 (4.0)	0 (0.0)	1 (0.9)	1 (0.8)	3 (1.1)
Psychiatric disorders, n (%)^a	0 (0.0)	3 (13.0)	11 (10.4)	12 (9.8)	26 (9.4)
Agitation	0 (0.0)	0 (0.0)	5 (4.7)	6 (4.9)	11 (4.0)
Mild	0 (0.0)	0 (0.0)	5 (4.7)	6 (4.9)	11 (4.0)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Delirium	0 (0.0)	3 (13.0)	3 (2.8)	4 (3.3)	10 (3.6)
Mild	0 (0.0)	2 (8.7)	1 (0.9)	2 (1.6)	5 (1.8)
Moderate	0 (0.0)	1 (4.3)	2 (1.9)	1 (0.8)	4 (1.4)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Renal and urinary disorders, n (%)^a	0 (0.0)	1 (4.3)	15 (14.2)	13 (10.7)	29 (10.5)
Acute kidney injury	0 (0.0)	0 (0.0)	9 (8.5)	8 (6.6)	17 (6.2)
Mild	0 (0.0)	0 (0.0)	2 (1.9)	3 (2.5)	5 (1.8)
Moderate	0 (0.0)	0 (0.0)	5 (4.7)	4 (3.3)	9 (3.3)
Severe	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)

Haematuria	0 (0.0)	0 (0.0)	5 (4.7)	1 (0.8)	6 (2.2)
Mild	0 (0.0)	0 (0.0)	2 (1.9)	1 (0.8)	3 (1.1)
Moderate	0 (0.0)	0 (0.0)	3 (2.8)	0 (0.0)	3 (1.1)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Respiratory, thoracic and mediastinal disorders, n (%)^a	3 (12.0)	2 (8.7)	17 (16.0)	14 (11.5)	36 (13.0)
Aspiration	1 (4.0)	0 (0.0)	2 (1.9)	3 (2.5)	6 (2.2)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	2 (1.6)	3 (1.1)
Moderate	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.4)
Severe	1 (4.0)	0 (0.0)	1 (0.9)	0 (0.0)	2 (0.7)
Pleural effusion	0 (0.0)	1 (4.3)	3 (2.8)	2 (1.6)	6 (2.2)
Mild	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Moderate	0 (0.0)	1 (4.3)	1 (0.9)	2 (1.6)	4 (1.4)
Severe	0 (0.0)	0 (0.0)	2 (1.9)	0 (0.0)	2 (0.7)
Skin and subcutaneous tissue disorders, n (%)^a	2 (8.0)	1 (4.3)	7 (6.6)	8 (6.6)	18 (6.5)
Rash	1 (4.0)	0 (0.0)	2 (1.9)	5 (4.1)	8 (2.9)
Mild	0 (0.0)	0 (0.0)	1 (0.9)	5 (4.1)	6 (2.2)
Moderate	1 (4.0)	0 (0.0)	1 (0.9)	0 (0.0)	2 (0.7)
Severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Vascular disorders	4 (16.0)	8 (34.8)	29 (27.4)	32 (26.2)	73 (26.4)

Hypertension	3 (12.0)	4 (17.4)	23 (21.7)	25 (20.5)	55 (19.9)
Mild	1 (4.0)	1 (4.3)	14 (13.2)	12 (9.8)	28 (10.1)
Moderate	1 (4.0)	2 (8.7)	4 (3.8)	6 (4.9)	13 (4.7)
Severe	1 (4.0)	1 (4.3)	5 (4.7)	7 (5.7)	14 (5.1)
Hypotension	2 (8.0)	4 (17.4)	11 (10.4)	10 (8.2)	27 (9.8)
Mild	0 (0.0)	2 (8.7)	5 (4.7)	5 (4.1)	12 (4.3)
Moderate	1 (4.0)	0 (0.0)	4 (3.8)	1 (0.8)	6 (2.2)
Severe	1 (4.0)	2 (8.7)	2 (1.9)	4 (3.3)	9 (3.3)

Abbreviations: MALS, macrophage activation-like syndrome; PT, preferred term; rhIFN γ , recombinant human activated interferon-gamma; SOC, system organ class; TEAE, treatment-emergent adverse event.

Footnote: The non-serious TEAEs were classified by severity using Medra 27.0 for events captured during the 90 days of follow-up for the enrolled 276 patients. In the analysis, non-serious TEAEs are categorised by severity gradings (mild, moderate and severe). It is clarified that the total of each SOC refers to the number of patients and not to the cumulative number of events.

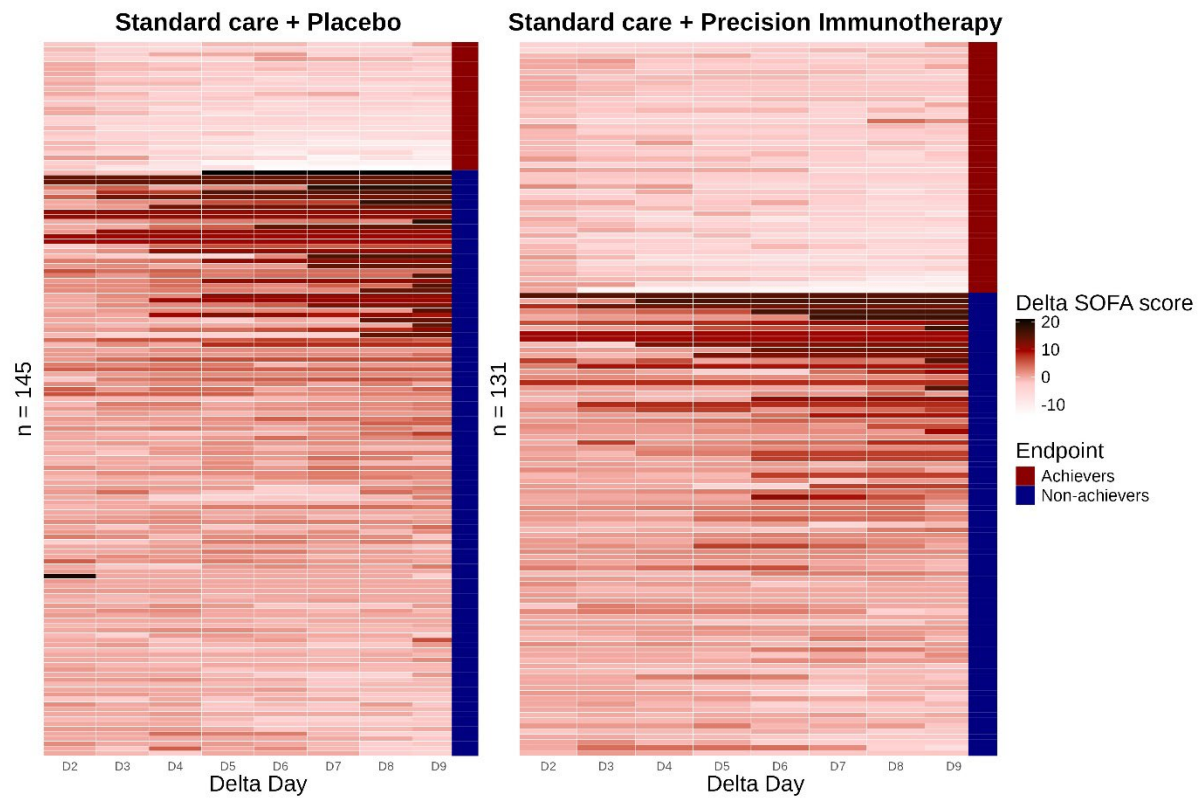
^aThese data represent the number of patients, whereas all other data in the table correspond to the number of events.

eTable 15. Quantification of the Benefit/Risk Ratio in the ImmunoSep Trial

Total enrolled patients		95% CIs
Number of patients needed to treat to attain the primary endpoint (NNT)	5.9	3.9 to 14.7
Number of patients needed to treat to induce at least 1 serious TEAE (NNH1)	200	-14.1 to 1220
Number of patients needed to treat to induce at least 2 serious TEAEs (NNH2)	6.6	3.0 to 27.8
Number of patients needed to treat to induce at least 3 serious TEAEs (NNH3)	10.3	4.7 to 50.0
Number of patients needed to treat to induce at least 4 serious TEAEs (NNH4)	7.7	4.18 to 58.8
Number of patients needed to treat to induce at least 5 serious TEAEs (NNH5)	12.5	5.4 to 37.0
Patients with macrophage activation-like syndrome		
Number of patients needed to treat to attain the primary endpoint (NNT)	3.3	1.9 to 25.6
Number of patients needed to treat to induce at least 1 serious TEAE (NNH1)	8.8	3.2 to 10.0
Number of patients needed to treat to induce at least 2 serious TEAEs (NNH2)	23.8	-5.7 to 33.3
Number of patients needed to treat to induce at least 3 serious TEAEs (NNH3)	10.9	-5.7 to 12.9
Number of patients needed to treat to induce at least 4 serious TEAEs (NNH4)	10.4	-5.9 to 12.9
Number of patients needed to treat to induce at least 5 serious TEAEs (NNH5)	10.1	-0.3 to 12.9
Patients with sepsis-induced immunoparalysis		
Number of patients needed to treat to attain the primary endpoint (NNT)	7.1	3.9 to 35.7
Number of patients needed to treat to induce at least 1 serious TEAE (NNH1)	47.6	9.9 to 55.8
Number of patients needed to treat to induce at least 2 serious TEAEs (NNH2)	5.6	3.4 to 17.9
Number of patients needed to treat to induce at least 3 serious TEAEs (NNH3)	9.9	3.9 to 76.2
Number of patients needed to treat to induce at least 4 serious TEAEs (NNH4)	7.2	5.9 to 23.8
Number of patients needed to treat to induce at least 5 serious TEAEs (NNH5)	13.7	5.9 to 23.8

Abbreviations: CI, confidence interval; NNT, number of patients needed to treat;

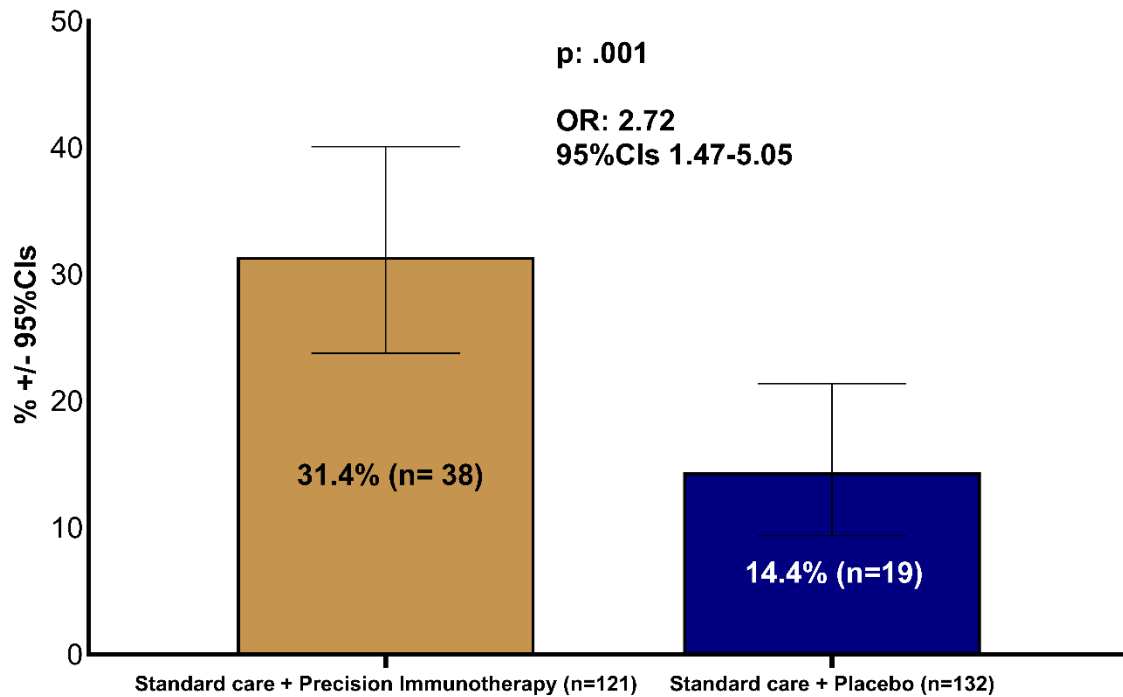
NNH, number of patients needed to harm.



eFigure 1. Heat-map of the changes of SOFA score over Follow-up the first 9 Days

Treatment started on day 1. The heat map shows the daily changes of SOFA score from baseline day 1.

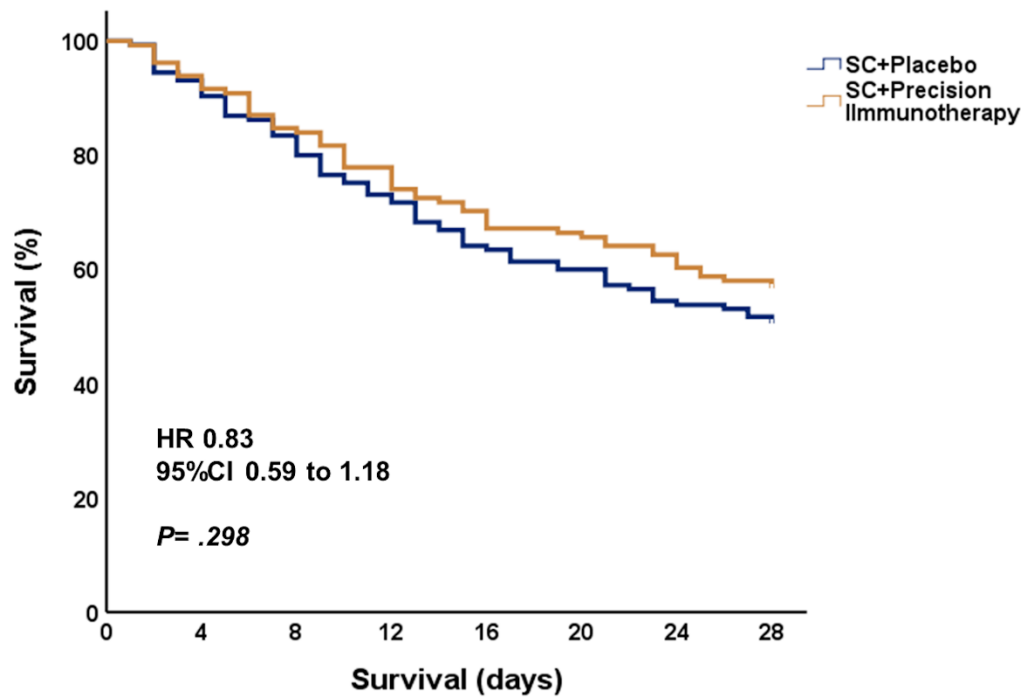
Abbreviations: n, number of participants per group; SOFA, sequential organ failure assessment.



eFigure 2. Sensitivity Analysis of the Attainment of the Primary Endpoint

Frequencies of patients experiencing at least 1.4 points decrease of mean SOFA score of days 2 to 9 from baseline. Patients who were discharged earlier than day 9 and for whom SOFA score were carried forward were excluded from the analysis.

Abbreviations CI, confidence interval; OR odds ratio; SOFA, sequential organ failure assessment.

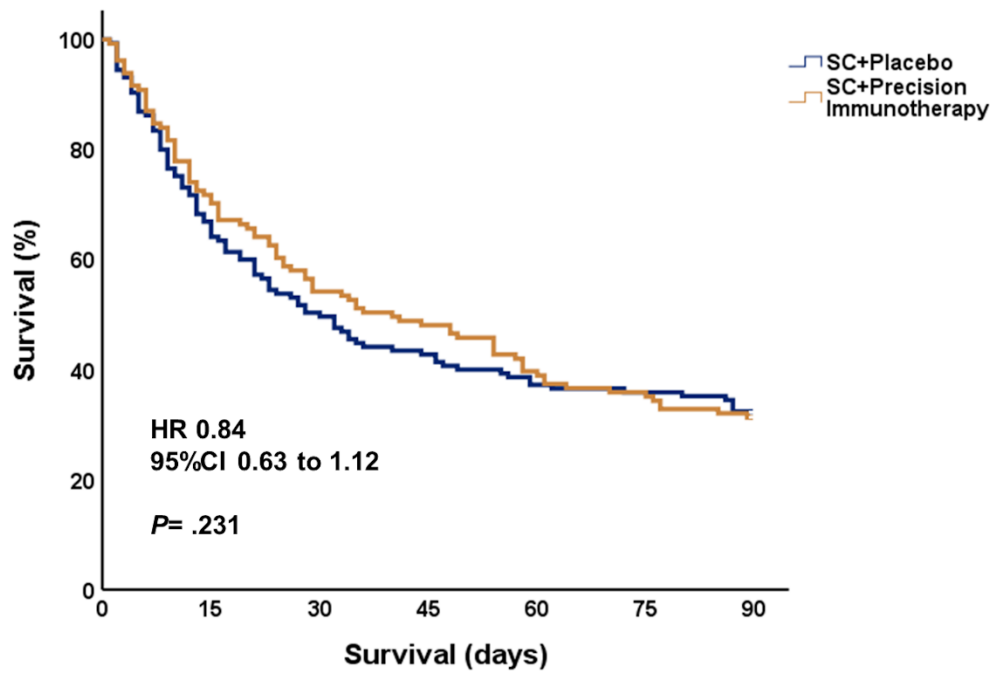


Patients at risk (n)

SC + Placebo	145	131	116	104	92	87	78	73
SC + Precision Immunotherapy	131	120	110	97	88	86	79	74

eFigure 3. Survival Analysis by Day 28

Abbreviations: CI, Confidence Interval; HR, Hazard Ratio; n, number of patients; SC, standard care.

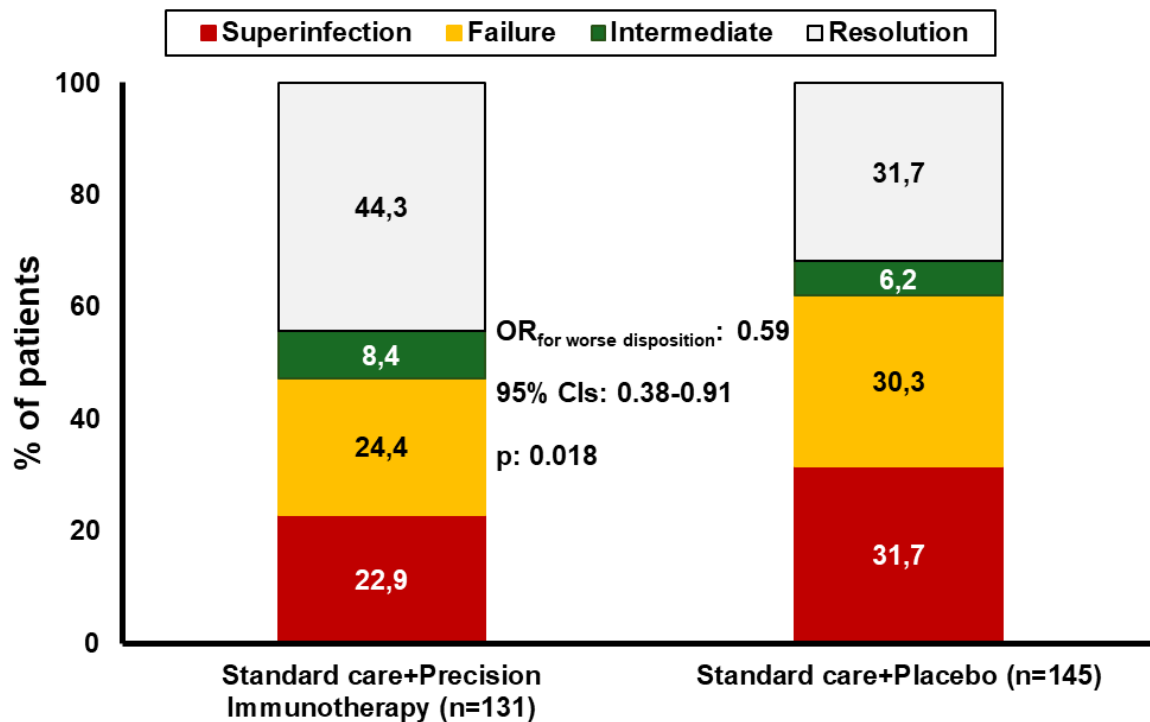


Patients at risk (n)

SC + Placebo	145	93	72	62	54	52	47
SC + Precision Immunotherapy	131	92	71	63	51	46	41

eFigure 4. Survival Analysis by Day 90

Abbreviations: CI, Confidence Interval; HR, Hazard Ratio; n, number of patients; SC, standard care.

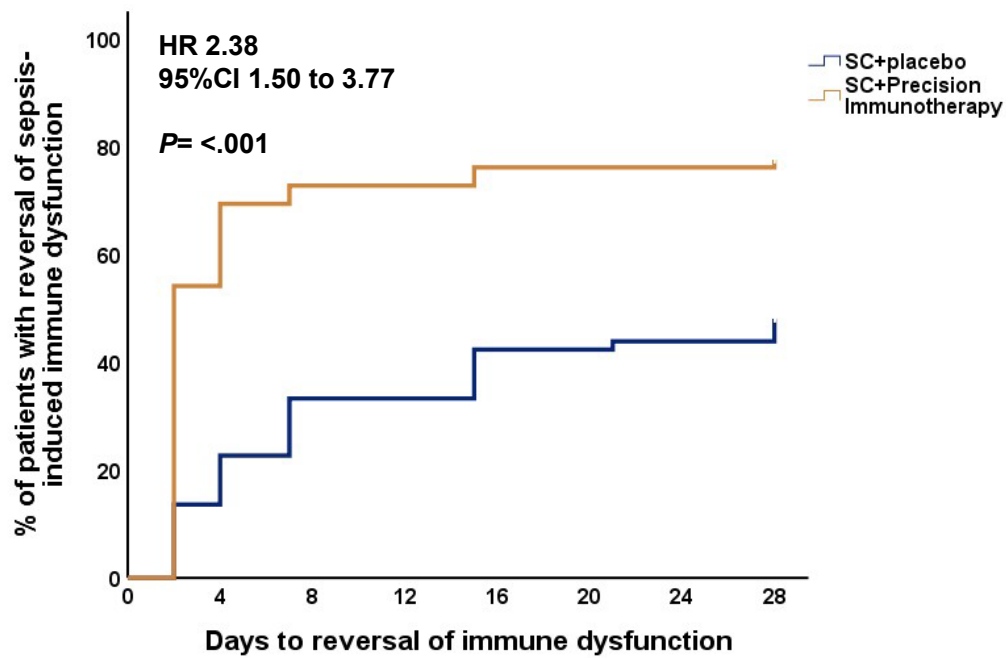


eFigure 5. Infection Disposition at Day 15

Abbreviations: CI, Confidence Interval; OR, odds ratio.

This endpoint applies on day 15 and the clinical state of the infection was considered as: resolved, according to the criteria for infection resolution provided in eTable 3; treatment failure, if infection persisted or relapsed; Superinfection, if a new infection due to a new pathogen emerged; and intermediate, if the outcome could not be classified in one of the above categories.

The OR for worse disposition/outcome includes superinfection, treatment failure and intermediate infection disposition



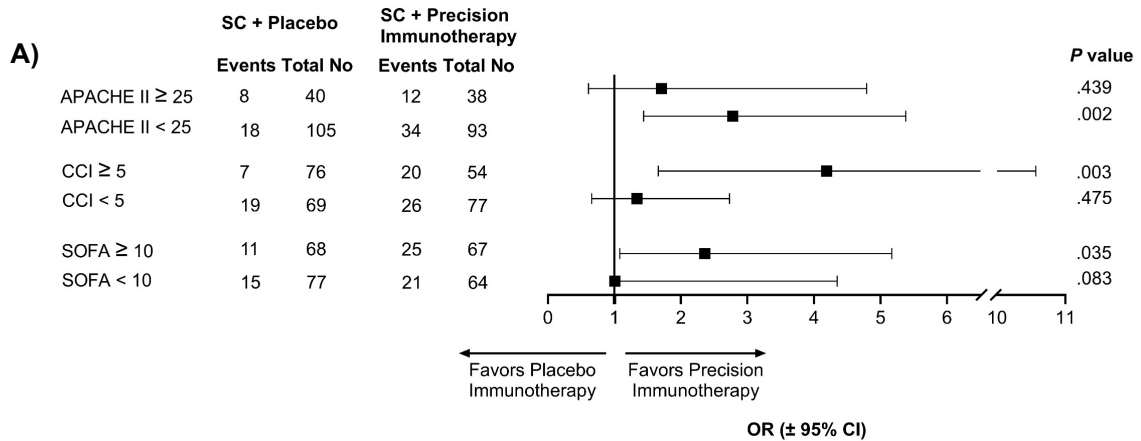
Patients at risk (n)

SC + Placebo	66	51	44	44	38	38	37	34
SC + Precision Immunotherapy	59	18	16	16	14	14	14	13

eFigure 6. Time to Reversal of Sepsis-Induced Immune Dysfunction

The reversal of immune dysfunction was studied in almost half of the studied patients, according to the study protocol. Groups were compared by Cox regression analysis and the HR was censored by Day 28. Blood measurements were done before start of the study drug and they were repeated on days 2, 4, 7, 15, 21 and 28. Missing draws were for patients dying or discharged earlier. This was defined as a) at least 15% decrease of ferritin for patients with MALS remaining decreased over follow-up time draws; and b) increase of the absolute number of HLA-DR receptors to more than 8,000 per CD45/CD14-monocyte for patients with SII remaining above these values over follow-up time draws. Patients not having at least two serial number of blood draws to confirm the permanence of the changes were considered not to attain reversal of sepsis-induced organ dysfunction.

Abbreviations CI, confidence interval; HLA-DR, Histocompatibility Complex DR; HR, hazard ratio; MALS, macrophage activation-like syndrome; n, number of patients; SC, standard care; SII, sepsis-induced immunoparalysis;



B)

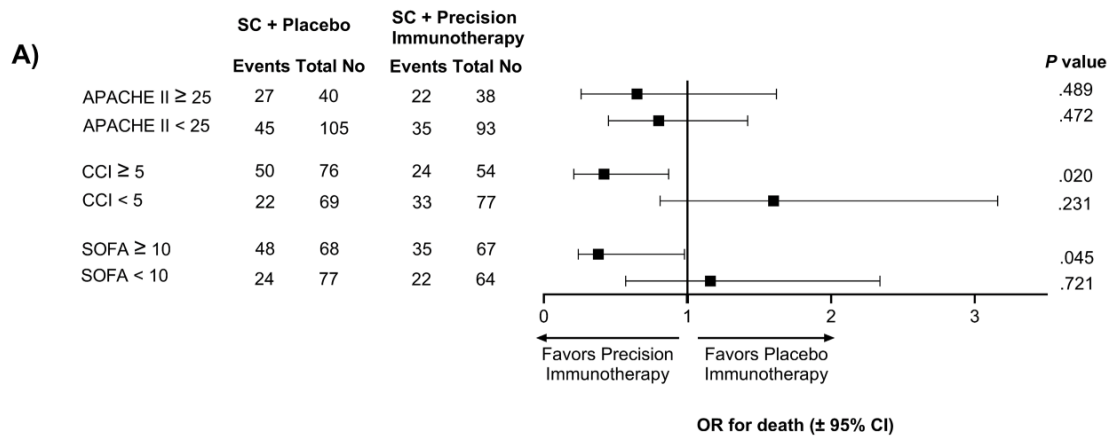
	OR	95% CIs	P-value
APACHE II ≥ 25	0.47	0.30-1.62	.70
APACHE II ≥ 25 x Precision Immunotherapy	1.85	0.66-5.19	.24
CCI ≥ 5	0.22	0.09-0.53	.001
CCI ≥ 5 x Precision Immunotherapy	5.79	2.34-15.05	<.0001
SOFA ≥ 10	0.56	0.27-1.19	.13
SOFA ≥ 10 x Precision Immunotherapy	3.08	1.37-6.96	.007

eFigure 7. Attainment of the Primary Endpoint According to Baseline Severity

A) According to the pre-defined statistical analysis plan, cut-offs of baseline APACHE, CCI and SOFA score were calculated which indicate increased risk for 28-day mortality. The odds for attainment of the primary endpoint (at least 1.4-point decrease of means SOFA of days 2 to 9 from baseline day 1) for each cut-off are provided.

B) Interaction tests of subgroups with the studied intervention for the primary endpoint.

Abbreviations: APACHE, acute physiology and chronic health evaluation; CCI, Charlson's comorbidity index; CI, confidence interval; OR, odds ratio; SC, standard care, SOFA, sequential organ failure assessment.



B)

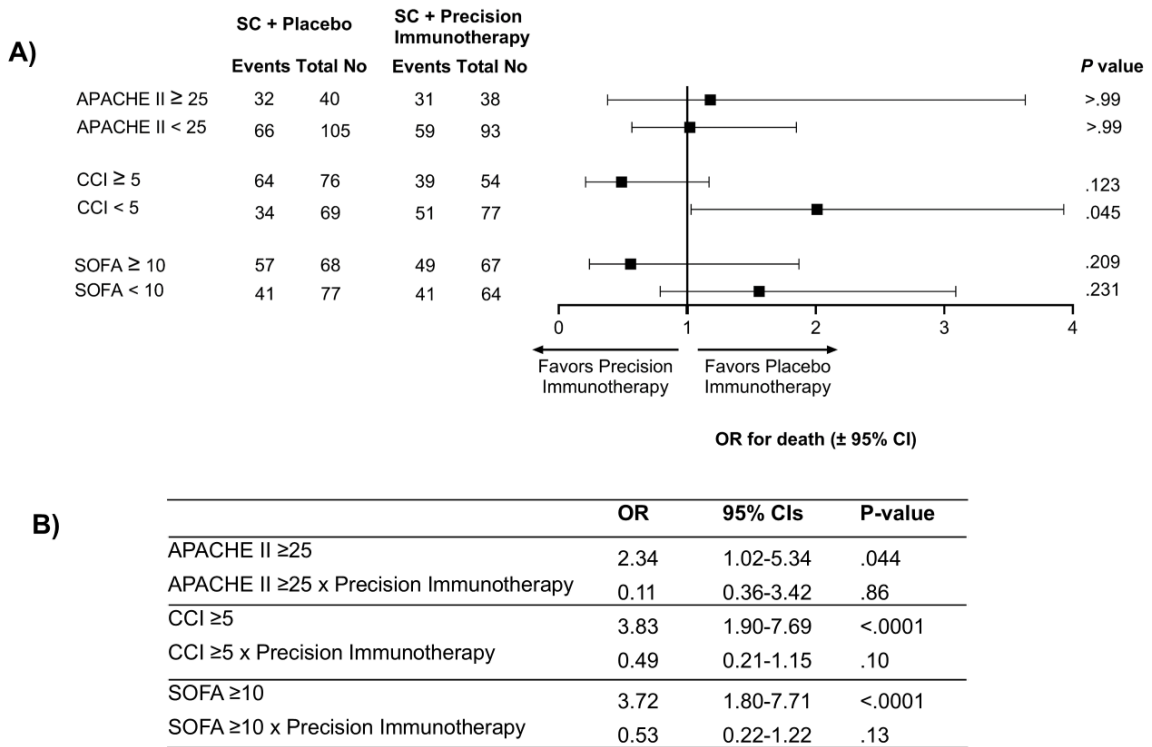
	OR	95% CIs	P-value
APACHE II ≥ 25	0.47	0.30-1.62	.70
APACHE II ≥ 25 x Precision Immunotherapy	1.85	0.66-5.19	.24
CCI ≥ 5	0.22	0.09-0.53	.001
CCI ≥ 5 x Precision Immunotherapy	5.79	2.34-15.05	<.0001
SOFA ≥ 10	0.56	0.27-1.19	.13
SOFA ≥ 10 x Precision Immunotherapy	3.08	1.37-6.96	.007

eFigure 8. 28-Day Mortality According to Baseline Severity

A) According to the pre-defined statistical analysis plan, cut-offs of baseline APACHE, CCI and SOFA score were calculated which indicate increased risk for 28-day mortality. The odds for death by each cut-off are provided.

B) Interaction tests of subgroups with the studied intervention for 28-day mortality.

Abbreviations: APACHE, acute physiology and chronic health evaluation; CCI, Charlson's comorbidity index; CI, confidence interval; OR, odds ratio; SC, standard care, SOFA, sequential organ failure assessment.



eFigure 9. 90-Day Mortality According to Baseline Severity

A) According to the pre-defined statistical analysis plan, cut-offs of baseline APACHE, CCI and SOFA score were calculated which indicate increased risk for 90-day mortality. The odds for death by each cut-off are provided.

B) Interaction tests of subgroups with the studied intervention for 90-day mortality.

Abbreviations: APACHE, acute physiology and chronic health evaluation; CCI, Charlson's comorbidity index; CI, confidence interval; OR, odds ratio; SC, standard care, SOFA, sequential organ failure assessment.

eReferences.

1. Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*. 2016;315(8):801-810. doi:10.1001/jama.2016.0287
2. Calandra T, Cohen J, International Sepsis Forum Definition of Infection in the ICU Consensus Conference. The international sepsis forum consensus conference on definitions of infection in the intensive care unit. *Crit Care Med*. 2005;33(7):1538-1548. doi:10.1097/01.ccm.0000168253.91200.83
3. Christ-Crain M, Jaccard-Stolz D, Bingisser R, et al. Effect of procalcitonin-guided treatment on antibiotic use and outcome in lower respiratory tract infections: cluster-randomised, single-blinded intervention trial. *Lancet*. 2004;363(9409):600-607. doi:10.1016/S0140-6736(04)15591-8
4. Kalil AC, Metersky ML, Klompas M, et al. Management of Adults With Hospital-acquired and Ventilator-associated Pneumonia: 2016 Clinical Practice Guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis*. 2016;63(5):e61-e111. doi:10.1093/cid/ciw353
5. Zilberberg MD, Shorr AF. Ventilator-associated pneumonia: the clinical pulmonary infection score as a surrogate for diagnostics and outcome. *Clin Infect Dis*. 2010;51 Suppl 1:S131-135. doi:10.1086/653062
6. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock: 2016. *Intensive Care Med*. 2017;43(3):304-377. doi:10.1007/s00134-017-4683-6